EXPERIMENT 5

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Experiment 5 – SQL Conditional Logic (Odd & Even Values)

Experiment 5: SQL Conditional Logic using the MOD (%) operator to classify employee salary values as Odd or Even. This experiment demonstrates numerical analysis and conditional evaluation using SQL queries in Oracle and PostgreSQL.

Aim
To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective
• To write and execute SQL queries using the MOD (%) operator.
• To retrieve salary details from an employee table.
• To classify salaries into Odd and Even categories.
• To display odd and even salary values separately.

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE)
PostgreSQL Database

Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

Practical / Experiment Steps

1. Create an employee table with columns:
   id
   name

department
salary

2. Insert sample employee salary records into the table.
3. Write a SQL query to retrieve employee salary details.
4. Use the MOD(salary, 2) function (Oracle) or salary % 2 (PostgreSQL) to check whether the salary is odd or even.
5. Use a CASE statement to classify salaries as:
   'Odd Salary'
   'Even Salary'
6. Execute separate queries to:
   Display only employees with Odd salaries.
   Display only employees with Even salaries.
7. Observe and record the output results.

Procedure

1. Open Oracle SQL Developer or pgAdmin.
2. Connect to the database.
3. Create the employee table using CREATE TABLE.
4. Insert sample records using INSERT INTO.
5. Execute SELECT queries with the MOD operator.
6. Apply CASE statement for conditional classification.
7. Display results and capture screenshots.

Input / Output Details

Input

Table Name: employee

Columns:
id (Integer)
name (Varchar)
department (Varchar)
salary (Integer)

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Logic Used:
MOD(salary, 2) or salary % 2
CASE statement

Step-wise Output

Step 1 – Create employee Table

```sql
CREATE TABLE Employees (
    employee_id INT PRIMARY KEY,
    name VARCHAR(50),
    salary INT
);

INSERT INTO Employees (employee_id, name, salary) VALUES
(1, 'Amit', 45000),
(2, 'Riya', 52001),
(3, 'Karan', 38000),
(4, 'Neha', 61015),
(5, 'Rahul', 50000);
```

Query result | Script output | DBMS output | Explain Plan | SQL histo

```
Table EMPLOYEES created.

Elapsed: 00:00:00.014
```

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Step 2 – Insert Data into employee Table

```sql
INSERT INTO Employees (employee_id, name, salary) VALUES
(1, 'Amit', 45000),
(2, 'Riya', 52001),
(3, 'Karan', 38000),
(4, 'Neha', 61015),
(5, 'Rahul', 50000);

SELECT employee_id,
       employee_name,
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
5 rows inserted.

Elapsed: 00:00:00.020
```

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Step 3 – Classify Salaries as Odd or Even

```sql
SELECT employee_id,
       name,
       salary,
       CASE
           WHEN MOD(salary, 2) = 0 THEN 'Even Salary'
           ELSE 'Odd Salary'
       END AS salary_type
FROM Employees;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.089 seconds

	EMPLOYEE_ID	NAME	SALARY	SALARY_TYPE
1	1	Amit	45000	Even Salary
2	2	Riya	52001	Odd Salary
3	3	Karan	38000	Even Salary
4	4	Neha	61015	Odd Salary
5	5	Rahul	50000	Even Salary

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Learning Outcome

After completing this experiment, the learner will be able to:

- Understand conditional logic in SQL queries.
- Use the MOD (%) operator for numerical analysis.
- Classify numeric values into odd and even categories.
- Write efficient SQL queries for payroll data analysis.
- Apply SQL logic in real-world enterprise environments.